

AHMED MOUNIR ALI

Backend Engineer

Alexandria, Egypt (Open to Relocation & Remote)

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Summary

Backend Engineer interested in **distributed systems, database internals, cloud-native architecture, and applied AI**. Experienced in building multi-tenant SaaS products, event-driven microservices, data pipelines, and performance-oriented systems using **Go, TypeScript/Node.js, Python, PostgreSQL, and Redis**. Enjoys translating complex requirements into reliable, maintainable backend services and understanding systems from first principles.

Technical Skills

Languages: Go, TypeScript/JavaScript, Python, PHP, SQL, Bash

Backend: Node.js, Express, NestJS, Laravel, REST APIs, WebSockets, Event-Driven Architecture (SQS)

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, pgvector; storage-engine internals (LSM-Trees, WAL, Bloom Filters)

Cloud & DevOps: Docker, AWS (ECS, ALB, RDS, S3, SQS, Lambda), Terraform, Linux, Prometheus & Grafana

AI & LLM Integration: RAG Pipelines, Vector Search (pgvector), Embeddings, LLM APIs (OpenAI, Anthropic Claude), LangChain

Frontend: React, Angular, Vue, Tailwind CSS (working knowledge)

Experience

Saspod

Backend Engineer (Contract)

April 2026 – Present

Scotland, UK (Remote)

- Building the core backend of a multi-tenant podcast SaaS platform using **TypeScript, Express.js, and PostgreSQL**, delivering production REST APIs through modular, transaction-safe service layers while collaborating asynchronously with a UK-based team.
- Designed a multi-tenant authentication and **RBAC** engine with **Redis**-backed refresh-token rotation and composable permission guards, enforcing tenant isolation across protected API resources and client workspaces.
- Shipped a secure user-invitation flow and admin client/studio management system, cutting client onboarding from a manual process to a fully automated **3-step** flow.
- Containerized services with **Docker** and built a custom job scheduler that automates daily purge workflows for data-retention compliance.

Projects

Global Opportunity Assistant (GOA) | Go, TypeScript, Python, PostgreSQL, pgvector, AWS

In Progress

- Led a **5-engineer team** and architected an end-to-end **9-service AI platform** in **Go/Fiber, NestJS, and FastAPI**, defining service boundaries, database ownership, versioned cross-language contracts, and **SQS event flows**.
- Designed the platform's **multi-agent AI architecture** across orchestration, RAG, writing, speaking, and ingestion workflows, using task-specific LLM agents, schema-validated outputs, bounded retries, and **Redis-backed token and cost controls**.
- Built a **Python web-scraping pipeline** with Playwright and BeautifulSoup to discover Engineering and Computer Science master's programs on MastersPortal, extract admission requirements, deduplicate records, and populate structured **RAG data**.
- Engineered a real-time **Go speaking-evaluation service** with a **4-agent IELTS scoring swarm**, running fluency, vocabulary, grammar, and pronunciation evaluations concurrently with partial-timeout handling and deterministic band aggregation.
- Built a **FastAPI/pgvector RAG service** with a **3-stage profile-matching pipeline**—hard eligibility gates, deterministic soft scoring, and LLM contextual boosts—plus Redis caching and **SQS-driven local profile replication**.

Chill-DB v2 | [\[GitHub\]](#) | Go, Bash

- Built a **custom LSM-Tree storage engine in pure Go** with a **WAL** (binary encoding + `fsync`) sustaining **0.6ms/insert** under full ACID crash durability guarantees.
- Reduced **hit latency by 80%** (3ms → 0.6ms) via Binary Search over sorted SSTables and **miss latency by 94%** (3ms → 0.18ms) via a Bloom Filter eliminating unnecessary disk I/O.
- Accelerated **compaction by 66%** (188ms → 63ms) via parallel SSTable merging; achieved **8 allocations/op** through zero-allocation write paths and buffer management.

HTTP/1.1 Server from Scratch | [\[GitHub\]](#) | *Go*

- Engineered a **zero-heap-allocation HTTP parser** using a cursor-based model over a pooled 64KB buffer, achieving **410 ns/op with 0 allocations** across all body sizes and header counts.
- Implemented a **bounded worker pool** (`runtime.NumCPU()` goroutines, 1000-slot queue) with **load shedding** (instant 503), ensuring the accept loop never blocks under any connection rate.
- Hardened against **Slowloris attacks** via a custom `idleTimeoutReader`; validated with a **fuzz harness over 2.8M+ mutations** with zero panics.

Point-of-Sale System | [\[GitHub\]](#) | *PHP, Laravel, Vue.js, Redis, Docker, AWS, Terraform*

- Designed and provisioned **multi-AZ** AWS cloud infrastructure across **2 Availability Zones** and **4 subnets** using **Terraform**, deploying a **3-stage** containerized architecture behind an **Application Load Balancer** with automated health checks.
- Architected a single-database multi-tenant POS engine with ORM-level scope isolation, utilizing **2 dedicated Redis databases** (DB0 for cache, DB1 for queues) to drive real-time analytics across **5 reporting modules**.
- Hardened API infrastructure by enforcing **3 tier sliding-window rate limiters** (throttled at 20–300 req/min), **transaction idempotency**, and **4 prioritized asynchronous queue networks** (*critical, default, notifications, reports*) managed by **Laravel Horizon**.
- Engineered an atomic, **transaction-safe** refund system to execute order reversals, automatically reconcile inventory stock levels across multi-location branches, and generate immutable audit log trails.
- Guaranteed application stability by authoring **117 automated unit and feature tests (722 assertions)** across **20 test suites** to validate rate-limiting policies, transaction idempotency, and multi-tenant boundary isolation.

Education

Information Technology Institute (ITI) <i>9-Month Professional Training Program — Open Source Application Development</i>	2025 – 2026 <i>Alexandria, Egypt</i>
B.Sc. in Electrical and Electronics Engineering <i>Alexandria University — CGPA: 3.24</i>	Sep. 2020 – June 2025 <i>Alexandria, Egypt</i>

Certifications & Achievements

- AWS Certified Solutions Architect Associate**
- AWS Certified Cloud Practitioner**
- Machine Learning Specialization** — DeepLearning.AI
- Deep Learning Specialization** — DeepLearning.AI
- IELTS Academic** — Band 8.0/9.0
- Competitive Programming** — Active LeetCode problem solver focused on algorithmic time-complexity optimization.

Volunteering

Media, IT and Marketing Member at EED Society	2022 – 2024
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